

Student Performance Analysis using NumPy, Pandas & Matplotlib

Learning Objectives

By the end of the project, students should be able to:

- Load datasets using Pandas
 - Explore and understand data
 - Clean and validate data
 - Calculate statistics using NumPy
 - Filter and sort data
 - Group and aggregate information
 - Discover patterns in the data
 - Create meaningful visualizations
 - Draw conclusions from the analysis
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Phase 1 – Import Libraries

Students will learn why each library is needed.

```
import numpy as np
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

Phase 2 – Load Dataset

```
df = pd.read_csv("students.csv")
```

Explain

- Rows
 - Columns
 - Dataset
 - DataFrame
-

Phase 3 – Explore the Dataset

Students should answer questions like

- How many students are there?
- How many columns?
- What are the column names?
- What are the data types?

Functions

`head()`

`tail()`

`shape`

`columns`

`dtypes`

`info()`

`describe()`

Phase 4 – Data Cleaning

Even if your dataset is already clean, intentionally introduce a few missing values before giving it to students so they can practice.

Topics

- Missing values

`isnull()`

`sum()`

- Fill missing values

`fillna()`

- Remove missing rows

`dropna()`

- Duplicate records

`duplicated()`

`drop_duplicates()`

Phase 5 – NumPy Statistics

Teach descriptive statistics.

`np.mean()`

`np.median()`

`np.max()`

`np.min()`

`np.std()`

`np.var()`

Questions

- Average exam score
 - Highest score
 - Lowest score
 - Average attendance
 - Average study hours
 - Average sleep hours
-

Phase 6 – Filtering Data

Students should find

Students scoring above 80

```
df[df["exam_score"]>80]
```

Students studying more than 6 hours

Students with attendance above 90%

Students sleeping less than 6 hours

Students with previous score greater than 75

Phase 7 – Sorting

Highest scorer

```
sort_values()
```

Sort by

- Exam score
 - Attendance
 - Study hours
-

Phase 8 – Create New Columns

Very important topic.

Example

Performance

```
if score >=80:
```

```
    Excellent
```

```
elif score>=60:
```

Good

else:

Needs Improvement

New column

Performance

Another column

Pass/Fail

Pass if Exam Score ≥ 35

Another column

Study Efficiency

Exam Score / Hours Studied

Phase 9 – Correlation Analysis

Ask students

Does studying more hours improve marks?

`corr()`

Columns

- Hours Studied
- Attendance
- Previous Scores
- Sleep Hours
- Exam Score

Discuss positive, negative, and weak relationships.

Phase 10 – Grouping

Group students by performance.

Example

Excellent

Good

Average

Poor

Find

- Average study hours
- Average attendance
- Average sleep
- Average exam score

using

groupby()

mean()

Phase 11 – Visualizations

This is where students bring the analysis to life.

1. Histogram

Distribution of Exam Scores

plt.hist()

Students learn

- Most common score range
 - Score spread
-

2. Bar Chart

Average study hours vs average exam score

3. Pie Chart

Pass vs Fail

4. Scatter Plot

Hours Studied vs Exam Score

This is probably the most important visualization.

Students immediately understand the relationship.

5. Scatter Plot

Attendance vs Exam Score

6. Line Chart

Sort students by previous scores

Plot

Previous Score

vs

Exam Score

Students can compare improvement.

7. Histogram

Sleep Hours

8. Box Plot (Optional)

Exam Score Distribution

Good introduction to outliers.

Phase 12 – Business Insights

Teach students that data analysis is about answering questions, not just creating charts.

Example insights:

- Average exam score of the class
 - Highest-performing student
 - Lowest-performing student
 - Average study hours
 - Students with attendance below 75%
 - Students studying more than 8 hours
 - Students sleeping less than 6 hours
 - Correlation between study hours and exam scores
 - Correlation between attendance and exam scores
 - Previous scores as a predictor of current performance
 - Percentage of students who passed
 - Percentage of students needing improvement
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Phase 13 – Final Dashboard

Bring all the charts together in one notebook with clear headings and short interpretations after each visualization. Students should present:

- Dataset overview
- Key statistics
- Cleaned data
- Visualizations
- Correlation findings
- Top-performing students
- Low-performing students

- Final conclusions and recommendations
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Suggested Student Tasks

1. Load and inspect the dataset.
2. Check for missing values and duplicates.
3. Calculate summary statistics using NumPy.
4. Identify the top 10 students by exam score.
5. Find students with attendance below 75%.
6. Create **Performance** and **Pass/Fail** columns.
7. Analyze the correlation between study hours, attendance, sleep, previous scores, and exam scores.
8. Create at least six visualizations (histogram, bar chart, pie chart, scatter plot, line chart, and box plot).
9. Write at least five insights based on the analysis.
10. Summarize the overall findings and suggest actions that could help improve student performance.